

Correction times for linear policy gradient on a coin line

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Abstract

How long does a rare corrective training signal take to overturn a learned preference for moving right? We answer this question for a linear logistic policy on a finite coin line. From zero initialization, the first correction time under exact return-gradient flow is of order $\varepsilon^{-\alpha_L}$, where ε is the training diversity and the explicit exponent $\alpha_L > 1$ depends on the line length and its parity. We first determine the zero-diversity trajectory up to a convergent offset. We then prove a perturbation bound up to the critical timescale and control the subsequent transition through the change of dominant failed paths. Numerical experiments illustrate the asymptotics; they are not used in the proofs.

1 Model and results

An agent moves on sites $0, \dots, L$, where $L \geq 4$, and succeeds on reaching a coin. The initial site is uniform. Each episode stops on its first success, including at time zero, or after $H = 2L$ actions. Actions move one site left or right, clipped at the endpoints. At position p with coin c , the policy moves right with probability

$$\sigma(z_{pc}), \quad z_{pc} = a + bp + dc, \quad \sigma(z) = (1 + e^{-z})^{-1}.$$

During training, $c \in \{1, \dots, L\}$ has law $\mu_c = (1 - \varepsilon)\mathbf{1}_{c=L} + \varepsilon/L$. Let $J_c(w)$ be the conditional success probability, for $w = (a, b, d)$, and set

$$J_\varepsilon = (1 - \varepsilon)J_L + \frac{\varepsilon}{L} \sum_{c=1}^L J_c, \quad w'_\varepsilon = \nabla J_\varepsilon(w_\varepsilon), \quad w_\varepsilon(0) = 0.$$

This is Euclidean gradient ascent on the exact return. For every training diversity the flow is globally defined: the return is a finite polynomial in logistic probabilities of linear forms, with bounded gradient and Hessian. The model and horizon conventions are those of MAIS-A8 [1].

At the test state the agent is at $k = \lceil L/2 \rceil$ and the coin is at zero. The logit $P_\varepsilon = a_\varepsilon + kb_\varepsilon$ is positive when the agent favors moving away from the coin. Define its first correction time by

$$T_\varepsilon = \inf\{t > 0 : P_\varepsilon(t) < 0\}, \quad \inf \emptyset = +\infty.$$

This paper quantifies first correction from zero initialization. The earlier O87 note [2] concerns eventual correction from every finite initialization, without a rate. First correction and permanent correction are different observables.

Write

$$n = \lceil (L + 1)/2 \rceil, \quad r = L + 1 - n, \quad u = (n, r, nL), \quad M = \|u\|^2, \quad x = (1, 2, 1),$$

$$\beta = \frac{x \cdot u}{M} = \frac{n(L+1) + 2r}{n^2(1+L^2) + r^2}, \quad \alpha = (1 - \beta)^{-1}.$$

The coefficient β comes from the corrective training state $(2, 1)$, whose feature vector is x ; the test state is $(k, 0)$. Here $r = n$ for odd L and $r = n - 1$ for even L . Since $r \leq n$ and $n \geq 3$, we have $0 < \beta < 1$ and $\alpha > 1$.

Theorem 1. *For every fixed integer $L \geq 4$, there are positive constants c_-, c_+, ε_0 such that*

$$c_- \varepsilon^{-\alpha} \leq T_\varepsilon \leq c_+ \varepsilon^{-\alpha} \quad (0 < \varepsilon \leq \varepsilon_0).$$

In particular, $\log T_\varepsilon / \log(1/\varepsilon) \rightarrow \alpha$ as $\varepsilon \downarrow 0$.

Thus correction takes asymptotically longer than $1/\varepsilon$. For $L = 4, 5, 6$, the exponents are $157/138$, $81/73$, and $601/567$, respectively. The theorem does not assert convergence of $\varepsilon^\alpha T_\varepsilon$ to a constant.

The proof has three parts. First we identify the logarithmic trajectory when all training coins are at L . Then we bound its perturbation under small diversity, giving the lower bound. For the upper bound, we keep a finite sum of minimal failed-path scores rather than following just one score through the transition. A ratio barrier makes the probe decrease while preserving the region in which that approximation is valid.

2 The zero-diversity trajectory

Set

$$C_L = \begin{cases} (L+1)^{-1}, & L \text{ odd}, \\ n(L+1)^{-1}, & L \text{ even}. \end{cases}$$

Theorem 2. *There is $w_\infty \in \mathbb{R}^3$, with $u \cdot w_\infty = \log(C_L M)$, such that*

$$w_0(t) = \frac{u}{M} \log t + w_\infty + o(1), \quad 1 - J_L(w_0(t)) \sim \frac{1}{Mt}. \quad (1)$$

Also $tw'_0(t) \rightarrow u/M$.

For this section only, write $w = w_0$, J_L for its return, and $z_p = z_{pL}$.

2.1 The cone containing the trajectory

For the moment, regard the nonterminal logits as independent. Their success derivatives $\kappa_p = \partial J_L / \partial z_p$ are strictly positive. Here is a direct justification. Let $V_j(p)$ be the probability of hitting L within j actions and let $\rho_i(p)$ be the probability of occupying p at time i before stopping. Differentiating the finite Bellman recursion gives

$$\kappa_p = \sigma(z_p) \sigma(-z_p) \sum_{i=0}^{H-1} \rho_i(p) [V_{H-i-1}(p+1) - V_{H-i-1}(\max(p-1, 0))].$$

Every bracket is nonnegative: from the farther site a path must first hit the nearer site, and reducing the remaining horizon cannot improve success. For strictness, put $j = H-1$, $p^- = \max(p-1, 0)$, and let τ be the first visit to $p+1$ starting at p^- . First-passage decomposition and horizon monotonicity give

$$V_j(p^-) \leq \Pr_{p^-}(\tau \leq j) V_j(p+1) < V_j(p+1).$$

The last inequality is strict because taking left at every step avoids $p + 1$ with positive probability, whereas $V_j(p + 1) > 0$ by taking right until success. Since $\rho_0(p) = 1/(L + 1)$, strict positivity follows.

The chain rule now gives

$$a' = \sum_{p < L} \kappa_p > 0, \quad b' = \sum_{p < L} p \kappa_p > 0, \quad d' = L a'.$$

Thus, along the trajectory,

$$d = L a, \quad 0 \leq b \leq (L - 1) a, \quad h := a + L d = (1 + L^2) a. \quad (2)$$

In particular $h - b \geq (L^2 - L + 2) a$. The flow exists for all time because its vector field is bounded and globally Lipschitz: the return is a finite polynomial in logistic probabilities of linear forms, whose first and second derivatives are bounded. Also $a(t) \rightarrow \infty$. Otherwise (2) and monotonicity would make all coordinates converge to a finite point, at which $a' > 0$. Continuity would then contradict convergence of a .

2.2 Failed paths and their scores

Let Ω consist of all pairs of a nonterminal starting site and a length- H action sequence that never reaches L . For $\omega \in \Omega$, let N count its left actions, let N_0 count left actions taken at zero, and let S be the sum of the sites at which left actions are taken. Write p_j for its position before action j and set

$$v_\omega = (N, S, L N), \quad x_p = (1, p, L).$$

Its conditional probability, given its starting site, is exactly

$$P_\omega(w) = e^{-v_\omega \cdot w} D_\omega(w), \quad D_\omega(w) = \prod_{j=0}^{H-1} (1 + e^{-z_{p_j}})^{-1}. \quad (3)$$

Consequently $F := 1 - J_L = (L + 1)^{-1} \sum_\omega P_\omega$ and

$$w' = \frac{1}{L + 1} \sum_\omega P_\omega(v_\omega - E_\omega), \quad E_\omega = \sum_{j=0}^{H-1} \sigma(-z_{p_j}) x_{p_j}. \quad (4)$$

On the cone (2), $z_p \geq h$, so, uniformly over this finite path set,

$$D_\omega = 1 + O(e^{-h}), \quad \|E_\omega\| = O(e^{-h}), \quad D_\omega \geq 2^{-H}. \quad (5)$$

All constants below may depend on L .

There are no clipped right actions on a failed path. Hence

$$p_H = p_0 + 2L - 2N + N_0 \leq L - 1.$$

It follows that $N \geq n$ and

$$S \geq N - N_0 \geq L + 1 + p_0 - N \geq L + 1 - N.$$

Since $n + r = L + 1$, the score difference satisfies

$$(v_\omega - u) \cdot w = (N - n)(h - b) + (N + S - n - r)b \geq 0. \quad (6)$$

Both coefficients on the right are nonnegative integers.

The score $(N, S) = (n, r)$ is attained, and its multiplicity is explicit. If L is odd, $r = n$ and the endpoint inequality forces $p_0 = N_0 = 0$. All left actions must be taken at site one: a left at site two or higher would make $S > n$. Once the walk reaches site two or higher, it cannot take another left action without violating this restriction. Thus all left actions precede a final monotone right run. The unique path consists of n right-left cycles from zero followed by right actions to site $L - 1$. If L is even, $r = n - 1$. The inequality $S \geq N - N_0$ forces $N_0 \geq 1$, and the endpoint inequality forces $N_0 = 1$ and $p_0 = 0$. Every other left action is at site one, since a left at site two or higher would make $S > n - 1$. Again, reaching site two forces the remaining actions to be right, so these possibilities are exhaustive. There are n paths: place the one left self-loop in any of the n slots before, between, or after the $n - 1$ right-left cycles, then finish with right actions. This proves the coefficient C_L in the theorem.

2.3 Separation of the dominant paths

To use (6), we must show that b also diverges. Let $F_{>}$ be the contribution to F from paths with $N > n$. Each has a score gap at least $h - b$. Comparing with one path of score u , using (3)–(5), gives

$$\frac{F_{>}}{F} = O(e^{-(h-b)}) \rightarrow 0. \quad (7)$$

For the remaining paths, $S \geq r$. The b component of (4), together with $S \geq 0$ on all paths, therefore gives

$$b' \geq r(F - F_{>}) - H(L - 1)e^{-h}F \geq \frac{r}{2}F$$

for sufficiently large time. The a component gives $0 < a' \leq HF$. Thus $b' \geq (r/(2H))a'$ eventually. After integration,

$$b \geq \frac{r}{2H}a - O(1).$$

In particular, $b \rightarrow \infty$ and $b \geq (r/(4H))a$ eventually.

For a path with $v_\omega \neq u$, at least one of the integer coefficients in (6) is positive. The cone bound and the preceding estimate therefore give a constant $\gamma > 0$ such that its score gap is at least γa at sufficiently late times. Summing the finite path set in (3) and (4), and using (5), yields

$$F = C_L e^{-U} (1 + O(e^{-\gamma a})), \quad (8)$$

$$w' = C_L e^{-U} (u + O(e^{-\gamma a})), \quad U = u \cdot w. \quad (9)$$

Here and below γ may be reduced to absorb the denominator errors. The error in the second line is a vector error in Euclidean norm.

2.4 Integrating the asymptotic flow

Since $a' > 0$ and $a \rightarrow \infty$, we may use a as the independent variable. Equation (9) gives

$$\frac{dw}{da} = \frac{u}{n} + O(e^{-\gamma a}).$$

The error is integrable. There is therefore a vector v such that

$$w = \frac{u}{n}a + v + O(e^{-\gamma a}). \quad (10)$$

Taking the inner product of (9) with u gives

$$\frac{d}{dt}e^U = C_L M(1 + O(e^{-\gamma a})) \longrightarrow C_L M.$$

It follows by integration that $e^U/t \rightarrow C_L M$, and hence

$$U = \log t + \log(C_L M) + o(1).$$

Combining this with (10) proves

$$w = \frac{u}{M} \log t + \left[v + \frac{u}{M} (\log(C_L M) - u \cdot v) \right] + o(1).$$

The bracket is w_∞ and has the required inner product with u . Finally, (8) and (9) give the return and velocity limits in Theorem 2.

3 Tracking and the lower bound

Theorem 3. *There are $K, c, \varepsilon_0 > 0$ such that for $0 < \varepsilon \leq \varepsilon_0$,*

$$\|w_\varepsilon(t) - w_0(t)\| \leq K\varepsilon((1+t)^{1-\beta} - 1), \quad 0 \leq t \leq c\varepsilon^{-\alpha}. \quad (11)$$

The constants can be chosen so that $P_\varepsilon(t) > 0$ for every $0 < t \leq c\varepsilon^{-\alpha}$.

3.1 The diversity perturbation

Write

$$\nabla J_\varepsilon = \nabla J_L + \varepsilon G, \quad G = \frac{1}{L} \sum_{c=1}^L \nabla J_c - \nabla J_L.$$

Fix a radius $R > 0$. All estimates in this section are uniform for $\|w - w_0(t)\| \leq R$, with constants allowed to depend on L, R . By (1) and boundedness on finite time intervals,

$$z_{pc}(w) = \frac{n + pr + cnL}{M} \log(1+t) + O_R(1) \quad (0 \leq p \leq L, 1 \leq c \leq L). \quad (12)$$

Define

$$\theta_c = \frac{n + cnL}{M}, \quad \beta_c = \frac{n + (c+1)r + cnL}{M}.$$

Thus $\theta_c \geq \theta_1$ and $\beta_c \geq \beta_1 = \beta$.

Lemma 1. *For a fixed coin, any event consisting of stopped action paths of at most H actions has at most $(L+1)2^H$ paths, including their starting sites. Each path probability p_ω satisfies*

$$\|\nabla p_\omega\| \leq H\sqrt{1 + 2L^2} p_\omega.$$

Thus a common bound on these probabilities bounds the gradient of the event probability, up to a constant depending only on L .

Proof. The paths are prefix-free at each starting site and may be injected into length- H action words. An action score is either $\sigma(-z_{pc})(1, p, c)$ or $-\sigma(z_{pc})(1, p, c)$; summing at most H such scores gives the displayed bound. Sum over the finite event. $\square$

For a start below c , differentiate failure rather than success. Every failed path stays below c . If it has N_- left actions, clipping at zero gives $p_H \geq p_0 + 2L - 2N_-$. Since $p_H \leq c - 1$,

$$N_- \geq \left\lceil \frac{2L + p_0 - c + 1}{2} \right\rceil \geq n.$$

By (12), each left-action probability below c is at most $C_R(1+t)^{-\theta_c}$. The failed-path probability and its summed gradient are therefore $O_R((1+t)^{-n\theta_c})$.

For a start above c , differentiate success. Every successful path includes a left action from $c+1$. Its probability is at most $C_R(1+t)^{-\beta_c}$ by (12); all other action factors are at most one. Summing the finite set of stopped successful paths gives a gradient bound $O_R((1+t)^{-\beta_c})$. There is no such starting site when $c = L$, and a start at c has zero derivative.

Finally,

$$n\theta_1 - \beta = \frac{n(n-1)(L+1) - 2r}{M} > 0.$$

Both parts of every conditional-return gradient are thus bounded by $C_R(1+t)^{-\beta}$. We have proved

$$\|G(w)\| \leq C_R(1+t)^{-\beta} \quad \text{if } \|w - w_0(t)\| \leq R. \quad (13)$$

3.2 An integrable one-sided Hessian bound

A norm bound on the entire Hessian would discard the sign of its leading term. We retain that sign using the failed-path representation for coin L . For each failed path ω , let P_ω be its conditional probability given its starting site, and put $g_\omega = \nabla \log P_\omega$. Then

$$\nabla^2 \log P_\omega = -B_\omega, \quad B_\omega = \sum_j \sigma(z_{p_j, L}) \sigma(-z_{p_j, L}) x_{p_j, L} x_{p_j, L}^\top,$$

where $x_{p, L} = (1, p, L)$ and the sum has H terms. Since $1 - J_L = (L+1)^{-1} \sum_\omega P_\omega$,

$$\nabla^2 J_L = \frac{1}{L+1} \sum_\omega P_\omega (B_\omega - g_\omega g_\omega^\top) \preceq \frac{1}{L+1} \sum_\omega P_\omega B_\omega. \quad (14)$$

The ordering is that of symmetric quadratic forms.

Set $\kappa = n(1+L^2)/M > 0$. Equation (12) gives $z_{p, L} \geq \kappa \log(1+t) - C_R$, and hence $B_\omega \preceq C_R(1+t)^{-\kappa} I$. Also the bound on path scores gives

$$P_\omega(w) \leq e^{RH\sqrt{1+2L^2}} P_\omega(w_0(t)).$$

Summing and using (1), we obtain $1 - J_L(w) \leq C_R(1+t)^{-1}$. It follows from (14) that

$$\nabla^2 J_L(w) \preceq K_R(t) I, \quad K_R(t) = C_R(1+t)^{-1-\kappa}, \quad \int_0^\infty K_R(t) dt < \infty. \quad (15)$$

These bounds hold for all $t \geq 0$ after increasing the constants on a compact initial interval. In particular, no assumption about the perturbed trajectory remaining in the zero-diversity invariant cone is needed.

3.3 Tracking and the first crossing

Let $\Delta(t) = \|w_\varepsilon(t) - w_0(t)\|$ and stop the comparison if Δ first reaches R . The line segment between the two parameters lies in the same radius- R ball. Integrating (15) along that segment and using (13) gives, while $\Delta < R$,

$$\Delta' \leq K_R(t)\Delta + \varepsilon C_R(1+t)^{-\beta}.$$

At $\Delta = 0$ this is understood as an upper right derivative; equivalently, one may regularize the norm. Grönwall's inequality and integrability of K_R yield a finite constant A_R such that

$$\Delta(t) \leq A_R \varepsilon ((1+t)^{1-\beta} - 1). \quad (16)$$

Consider

$$S_\varepsilon = \left(1 + \frac{R}{2A_R \varepsilon}\right)^{1/(1-\beta)} - 1.$$

If the first exit occurred at or before S_ε , continuity and (16) would imply $R \leq R/2$. Thus there is no exit, and (16) holds through S_ε , and

$$S_\varepsilon \sim \left(\frac{R}{2A_R}\right)^\alpha \varepsilon^{-\alpha}.$$

To ensure a positive probe, put $v_* = (1, k, 0)$. Since $a'_0, b'_0 > 0$, P_0 is strictly increasing from zero. Choose

$$0 < R \leq \min \left\{ 1, \frac{P_0(1)}{2\|v_*\|} \right\}.$$

For $1 \leq t \leq S_\varepsilon$, the tracking bound $\Delta \leq R/2$ gives $P_\varepsilon(t) \geq P_0(1) - \|v_*\|R/2 > 0$. On $0 \leq t \leq 1$, let $m = \min_{[0,1]} P'_0 > 0$. Concavity of $(1+t)^{1-\beta}$ implies $\Delta(t) \leq A_R \varepsilon (1-\beta)t$, so

$$P_\varepsilon(t) \geq (m - \|v_*\|A_R \varepsilon(1-\beta))t > 0 \quad (0 < t \leq 1)$$

for sufficiently small ε . Reduce ε_L so that $S_\varepsilon \geq 1$, and choose $c_L > 0$ with $c_L \varepsilon^{-\alpha} \leq S_\varepsilon$ throughout the remaining range. Equation (16) and these positivity bounds prove Theorem 3.

4 The upper bound

Write $E = \log(1/\varepsilon)$ and $W = \alpha u/M$ throughout this section. We first find a region in which a finite failed-path sum gives a uniform approximation to the flow. We then prove that the trajectory enters that region with enough corrective drift to force a crossing.

4.1 An approximation through the sign change

Put

$$A = a + d - b, \quad D = d - a, \quad q = a + 2b + d = x \cdot w, \quad h = a + Ld.$$

Fix $\delta > 0$ smaller than half of both $A(W)$ and $D(W)$; these quantities are positive. All subsequent constants may depend on this choice. Consider the region

$$A \geq \delta E, \quad D \geq \delta E, \quad P \geq 0. \quad (17)$$

Every training logit in this region is at least δE . Indeed, if $b \geq 0$, then $d = (A + D + b)/2 > 0$ and $a + d = A + b \geq \delta E$. If $b < 0$, $a \geq -kb$, $d = a + D > 0$, and

$$a + Lb + d = 2a + Lb + D \geq (L - 2k)b + D \geq \delta E.$$

Also

$$b + d \geq \delta E. \quad (18)$$

For $b \geq 0$, use $b + d = (A + D + 3b)/2$; for $b < 0$, use $b + d = D + a + b \geq D + (1 - k)b$. In particular $q = z_{2,1} \geq \delta E$, since $(2, 1)$ is a training state. Equivalently, $q = A + 3b$ for $b \geq 0$, and $q = 2a + 2b + D \geq 2(1 - k)b + D$ for $b < 0$.

For each failed length- $2L$ path with coin L , let N be its number of left actions and S the sum of the sites at those actions. The path count in Section 2 gives

$$N \geq n, \quad N + S \geq L + 1 = n + r, \quad S \leq N(L - 1).$$

Let m_S be the number of such paths, including their starting sites, with $N = n$ and position sum S . Define the positive exponential sum

$$F(w) = \frac{e^{-nh}}{L + 1} \sum_{S=r}^{n(L-1)} m_S e^{-Sb}, \quad s(b) = \frac{\sum_S S m_S e^{-Sb}}{\sum_S m_S e^{-Sb}}, \quad z(b) = (n, s(b), nL). \quad (19)$$

Then $-\nabla F = Fz$ and $r \leq s(b) \leq n(L - 1)$. The endpoint counts m_r and $m_{n(L-1)}$ are positive, as justified below. The following path facts will also be used. From any start $p_0 < c$, a failed path stays below c and satisfies $p_0 + 2L - 2N_- \leq p_{2L} \leq c - 1$. Thus

$$N_- \geq \lceil (2L + p_0 - c + 1)/2 \rceil \geq n.$$

Every successful path starting above c must take a left action at $c + 1$. For $c = 1$, the only such path with one left action is the immediate step from site 2: any larger start or any preceding right action requires another left action before reaching the coin. The endpoint score $S = n(L - 1)$ is attained from start zero by climbing to $L - 1$, then taking $(\text{left}, \text{right})^n$ for odd L ; for even L , take $(\text{left}, \text{right})^{n-1}$ followed by one left. The lengths are respectively $L - 1 + 2n = 2L$ and $L - 1 + 2n - 1 = 2L$. Both paths fail. Positivity of m_r follows from the explicit paths used in the zero-diversity trajectory proof.

Set

$$Q(w) = \frac{\varepsilon}{L(L + 1)} e^{-q}. \quad (20)$$

Lemma 2. *Uniformly in (17),*

$$\nabla J_\varepsilon = Fz - Qx + \mathcal{E}, \quad \|\mathcal{E}\| \leq C(\varepsilon + e^{-\delta E})(F + Q). \quad (21)$$

Proof. For each fixed coin and starting site, success and failure probabilities sum to one. We may therefore differentiate success from above and the negative of failure from below. Starts at the coin have zero derivative. We treat the entire coin- L contribution first, and use this decomposition only for the remaining coins $1 \leq c < L$, so no term is counted twice.

A failed path for coin L has probability $e^{-Nh - Sb}$ times a product of $2L$ logistic denominators. Since every logit is at least δE , these denominators equal $1 + O(e^{-\delta E})$; their logarithmic gradients are $O(e^{-\delta E})$. These estimates are uniform over the finite path set.

Paths with $N > n$ have a score gap of at least δE relative to one of the endpoint scores in (19). For $b \geq 0$, write

$$Nh + Sb - (nh + rb) = (N - n)(h - b) + (N + S - n - r)b,$$

where $h - b = ((L + 1)A + (L - 1)D + (L - 1)b)/2 \geq \delta E$. For $b < 0$, write

$$Nh + Sb - n[h + (L - 1)b] = (N - n)[h + (L - 1)b] + [S - N(L - 1)]b,$$

where $h + (L-1)b = z_{L-1,L} \geq \delta E$. Thus their probabilities and gradient contributions are $O(e^{-\delta E} F)$. The coin- L term is $Fz + O((\varepsilon + e^{-\delta E})F)$ after its mixture weight is included.

For a start above coin 1, the unique successful path with only one left action is the immediate left step from site 2. Its gradient, with both initial-site and coin weights, is $-Qx + O(e^{-\delta E} Q)$. Every other successful path from above includes that left step and at least one further left action. Its probability is at most $e^{-q-\delta E}$. The norm of a path's logarithmic gradient is uniformly bounded, so its gradient obeys the same estimate up to a constant.

For coins $2 \leq c < L$, success from above requires a left action at $c+1$. Its logit is $q + (c-1)(b+d) \geq q + \delta E$, by (18). Finally, for the remaining coins $1 \leq c < L$, failure from below requires at least n left actions. If $b \geq 0$, its minimum below-coin logit is $a + dc \geq a + d = A + b$, and

$$n(a + d) - q = (n-1)A + (n-3)b \geq \delta E.$$

If $b < 0$, that minimum is $a + (c-1)b + dc = q + (c-1)(b+d) - 2b \geq q$, so n copies exceed q by at least δE . Differentiate these below-coin contributions through their failure probabilities. Summing all the finitely many paths proves the lemma. $\square$

4.2 A ratio barrier

Define $R = Q/F$ and $R_0 = nL/2$. For $z = (n, s, nL)$, set $M_s = \|z\|^2$ and $D_s = z \cdot x = n(L+1) + 2s$. The exact logarithmic derivatives of F, Q imply

$$(\log R)' = (z - x) \cdot \nabla J_\varepsilon = F\{M_s - D_s - (D_s - 6)R\} + (z - x) \cdot \mathcal{E}. \quad (22)$$

At $R = R_0$, the expression in braces is

$$s^2 - (nL + 2)s + n^2 \left(1 + \frac{L^2 - L}{2}\right) + n(2L - 1).$$

Its minimum over all real s is

$$n^2 \left(1 + \frac{L^2}{4} - \frac{L}{2}\right) + n(L - 1) - 1 > 0. \quad (23)$$

For sufficiently small ε , Lemma 2 therefore makes $R' > 0$ at $R = R_0$, throughout (17). A trajectory entering with $R > R_0$ cannot cross down through this barrier while in the region.

For $R \geq R_0$, we have

$$\frac{D_s}{R} - 6 \leq -\frac{2}{L}, \quad \frac{n + ks}{R} - (1 + 2k) \leq -1.$$

Using (21), and reducing the permitted ε if needed,

$$q' \leq -\frac{Q}{L}, \quad P' \leq -\frac{Q}{2}. \quad (24)$$

The two other linear coordinates satisfy

$$A' \geq -C\zeta Q, \quad D' \geq -C\zeta Q, \quad \zeta = \varepsilon + e^{-\delta E}. \quad (25)$$

Indeed, the $-Qx$ contribution is zero in both coordinates, and the Fz contributions are $F[n(L+1) - s] \geq 2nF$ and $Fn(L-1) > 0$. The error is $O(\zeta Q)$ since $F \leq Q/R_0$.

Suppose the flow enters at time t_e with

$$A, D \geq 2\delta E, \quad 0 < P \leq C_0 E, \quad R > R_0, \quad q \leq (\alpha - 1)E + C_0. \quad (26)$$

Define t^* to be the first time after t_e at which $A = \delta E$, $D = \delta E$, $R = R_0$, or $P = 0$. Before this time all preceding estimates hold. For any such t , (24) gives $\int_{t_e}^t Q du \leq 2P(t_e) \leq 2C_0 E$. By (25),

$$A(t), D(t) \geq 2\delta E - 2CC_0\zeta E > \delta E$$

for sufficiently small ε . These inequalities also hold at a finite t^* by continuity. The alternative $R(t^*) = R_0$ is impossible because $R' > 0$ there, whereas a first crossing from above would have $R' \leq 0$. Hence any finite t^* is a probe zero. If $t^* = \infty$, the same estimates hold for all later times, which the following time bound rules out. But (24) also gives

$$(e^q)' \leq -\frac{\varepsilon}{L^2(L+1)}.$$

A positive e^q cannot satisfy this inequality for more than $L^2(L+1)e^{q(t_e)}/\varepsilon = O(\varepsilon^{-\alpha})$ time. The probe must therefore reach zero within that time. Since $P' < 0$ at this first zero, it crosses into $P < 0$ immediately. It remains to establish the entrance conditions (26) at $t_e = O(\varepsilon^{-\alpha})$.

4.3 Reaching the ratio barrier

For y in a fixed compact set, $WE + y$ lies in the region (17) for all sufficiently large E : the forms A, D, P are strictly positive at W , and $A(W), D(W) > 2\delta$. Also $W_b > 0$, so the finite sum in (19) is uniformly dominated by its $S = r$ term. Consequently,

$$\varepsilon^{-\alpha} F(WE + y) \rightarrow C_L e^{-u \cdot y}, \quad s(b) \rightarrow r, \quad \varepsilon^{-\alpha} Q(WE + y) = B e^{-x \cdot y}, \quad B = \frac{1}{L(L+1)}.$$

Here $u \cdot W = \alpha$, $x \cdot W = \alpha - 1$, and $C_L = m_r/(L+1) > 0$ was computed in Section 2. Lemma 2 now gives the uniform compact-set limit

$$\varepsilon^{-\alpha} \nabla J_\varepsilon(WE + y) \rightarrow C_L u e^{-u \cdot y} - B x e^{-x \cdot y}. \quad (27)$$

Indeed, its rescaled error is bounded by a constant times $(\varepsilon + e^{-\delta E})\varepsilon^{-\alpha}(F + Q)$, which tends uniformly to zero. Thus all path classes are controlled by the same estimate used after entrance.

In time $\tau = \varepsilon^\alpha t$, the limiting equation for y is the right side of (27). Write $U = u \cdot y$, $q_y = x \cdot y$, $D_+ = u \cdot x$, and introduce a second clock by $d\xi/d\tau = C_L e^{-U}$. In this clock the limiting equation and ratio are

$$\frac{dy}{d\xi} = u - \mathcal{R}x, \quad \mathcal{R} = \frac{B}{C_L} e^{U - q_y}, \quad \frac{d\mathcal{R}}{d\xi} = \mathcal{R}\{(M - D_+) - (D_+ - 6)\mathcal{R}\}. \quad (28)$$

Since $M > D_+ > 6$, every positive ratio converges to $\mathcal{R}_* = (M - D_+)/(D_+ - 6)$. Applying (23) at $s = r$ shows $\mathcal{R}_* > R_0$.

Choose a fixed, sufficiently small $\tau_0 > 0$ so that (11) applies at $t_0 = \tau_0 \varepsilon^{-\alpha}$. Equations (1) and (11) put

$$y_\varepsilon(t_0) := w_\varepsilon(t_0) - WE$$

in a fixed compact set, independent of sufficiently small ε . On this set the initial limiting ratio is bounded above and bounded away from zero. Choose R_1 strictly between R_0 and $\mathcal{R}_*$. The logistic equation (28) then puts every such limiting solution above R_1 by a single finite clock time ξ_* .

Apply the same positive clock factor $C_L e^{-u \cdot y}$ to the rescaled full equations. Let $\mathcal{K}_0$ contain their initial points, and let $\mathcal{K}_1$ contain all limiting trajectories from $\mathcal{K}_0$ on $[0, \xi_*]$. The logistic ratio stays between its initial value and its positive equilibrium, uniformly over $\mathcal{K}_0$. Thus these orbits and their velocities are bounded and $\mathcal{K}_1$ is compact. On the closed one-neighbourhood $\mathcal{K}_2$ of $\mathcal{K}_1$, the clock factor is bounded above and away from zero. The transformed full fields converge uniformly to the smooth limiting field. If their uniform difference is $\eta_\varepsilon \rightarrow 0$ and a Lipschitz constant for the limit field on a containing compact ball is H_0 , comparison up to first exit gives

$$\sup_{0 \leq \xi \leq \xi_*} \|y_\varepsilon(\xi) - y(\xi)\| \leq \eta_\varepsilon \xi_* e^{H_0 \xi_*}.$$

For small ε this is less than one, excluding exit from $\mathcal{K}_2$. The initial limiting point is always chosen equal to the full initial point; convergence of the initial points themselves is not required.

The ratios Q/F converge uniformly on $\mathcal{K}_2$ to $\mathcal{R}$, since $W_b > 0$ makes the $S = r$ term dominate the finite sum. The comparison therefore gives $R > R_0$ at clock time ξ_* , uniformly over the allowed initial points. Finally $u \cdot y$ is bounded above on $\mathcal{K}_2$, so $d\tau/d\xi = e^{u \cdot y}/C_L$ has a uniform upper bound. The elapsed rescaled time is bounded independently of ε . We have therefore reached $t_e = O(\varepsilon^{-\alpha})$ with $w_\varepsilon(t_e) = WE + O(1)$ and $R > R_0$. The linear forms A, D, P are strictly positive at W , while $q(W) = \alpha - 1$. Recall that δ was fixed smaller than half of both $A(W)$ and $D(W)$. Increasing C_0 if necessary gives the remaining entrance bounds. This gives (26). If a probe crossing occurred earlier, the desired upper bound already holds. The preceding section proves the upper bound. Together with Theorem 3, this completes Theorem 1.

5 Numerical evidence and reproduction

The companion code evaluates the return and its gradient by finite-horizon dynamic programming. It propagates both success and failure probabilities and uses the less saturated value difference in the gradient recursion. Independent rational path enumeration checks the implementation, including a saturated-policy case. No Monte Carlo return estimates are used.

We integrated the flow from zero for $L = 4, 5, 6$ and $\varepsilon = 10^{-p}$, using

$$p \in \{2, 4, 8, 12, 16, 24, 32, 48, 64, 96\}.$$

All thirty runs reached a downward probe crossing. Integration used the clock $s = \log(1 + t)$ and DOP853 with relative tolerance 2×10^{-9} . Six repeats with RK45 and relative tolerance 2×10^{-11} changed crossing times by at most 2.20×10^{-10} in relative terms. Independent 180-digit differentiation at the three $\varepsilon = 10^{-96}$ crossing states agreed with the computed gradients to relative error below 4.6×10^{-14} . These checks are not certified integration bounds.

L	Theorem exponent	Secant slope, 10^{-64} to 10^{-96}
4	1.137681159	1.138261638
5	1.109589041	1.110163109
6	1.059964727	1.061551262

The companion README records the environment, commands, source snapshot, and result files. Exact finite checks cover the score inequalities, both endpoint path counts, the ratio-barrier identity, and the invariant-region algebra. The theoretical arguments apply to every fixed integer $L \geq 4$; the numerical checks cover only their stated finite ranges.

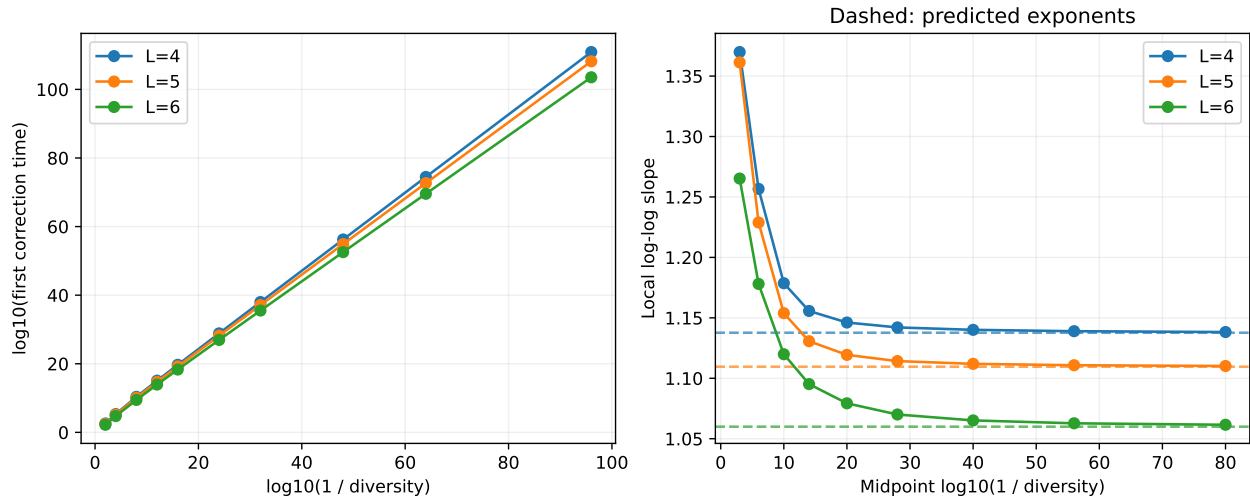


Figure 1: First correction times and successive log-log slopes from the retained numerical runs. Dashed lines show the theorem’s exponents. The finite-range slopes are evidence about convergence, not a proof of it.

6 Scope and remaining questions

The result concerns exact continuous-time return-gradient ascent, fixed L , and zero initialization. It supplies two-sided order bounds but no sharp prefactor. A limiting law for Gaussian initializations, the time of permanent correction, discrete gradient ascent, and sampled policy-gradient updates remain separate questions. Constants may depend on L ; no bound uniform as $L \rightarrow \infty$ is asserted. The prefactor suggested by a reduced transition ODE is exploratory and is not needed for the theorem.

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References

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